FILIPP LOTSMANOV

Applied Data Science & AI Student

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EDUCATION

Breda University of Applied Sciences Sep 2024 – Present
BSc Applied Data Science & AI · 8.5/10 (≈3.8 GPA) *Breda, Netherlands*

PROFILE

Third-year Applied Data Science & AI student on the Data Scientist and Data Engineer tracks. I train vision and sequence models, measure them on held-out data, and ship them behind real-time services.

TECHNICAL SKILLS

Languages: Python, Go, SQL
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Keras
Classical ML: scikit-learn, XGBoost, statsmodels
Computer vision: OpenCV, MediaPipe, torchvision, albumentations, U-Net
NLP & LLMs: SentencePiece/BPE, NLLB, faster-whisper, Amazon Bedrock, RAG, LangChain, FAISS
Optimisation & serving: TorchScript, quantisation, mixed-precision, FastAPI, Ollama, NVIDIA Jetson
MLOps & infra: MLflow, ClearML, Docker, Kubernetes, Azure, AWS, Terraform, GitHub Actions, Airflow, Prometheus
Data & analysis: PostgreSQL, pandas, NumPy, GeoPandas, Shapely, Power BI

PROJECTS & TECHNICAL WORK

Plant Segmentation MLOps Platform Feb – Jun 2026
Team of 5, BUas – NPEC

- Airflow orchestration and Azure ML pipelines behind a U-Net root-segmentation service: six DAGs for preprocessing, hyperparameter search and feedback-driven retraining with champion-challenger promotion.
- Owned most of the cv-pipeline package (inference, training, CLI) served through FastAPI; 0.8371 test F1 / 0.7199 IoU on 20,512 held-out patches, 85% coverage floor enforced in CI.

Video Emotion Classification Pipeline Sep – Nov 2025
Group Project – Media Client

- Nine-stage video-to-emotion-timeline system for an external client brief; fine-tuned DeBERTa-v3-base on 419,180 samples to 0.8162 macro F1 (0.9202 accuracy) on a 62,862-sample held-out set.
- Built a Transformer encoder-decoder from scratch; owned the transcription and scene-detection stages.

Grounded Q&A on AWS Bedrock Sep – Oct 2026
Personal Project

- Built a serverless RAG service over 10 research papers that cites a source for every answer, declines out-of-scope questions (7/7 in evaluation) and costs about $0.001 per question.
- Integrated JWT authentication (Cognito sign-in, PKCE), request throttling and email budget alerts.
- Built an evaluation harness for retrieval, refusals and citations, and stress-tested the LLM judge with seeded errors, documenting 3 wrong answers that every metric passed.

Concurrent ETL Pipeline in Go May – Aug 2026
Personal Project

- Streaming CSV-into-PostgreSQL pipeline: 2.8–5.2 MB heap across a 2.3 GB, 27.6M-record file at ∼41k rec/sec.
- 4.76× enrichment speedup on 14 cores via a goroutine worker pool; live-Postgres tests gate the Docker build and CI.

COMPETITIONS

BrabantHack 2026 – Winner, DEMCON Deep Tech Track Apr 2026
Team of 3

- Predicted off-screen pedestrian positions from shadow imagery alone: PyTorch ResNet-50 fused with 19 hand-crafted geometric features, iterating 0.52 → ∼0.62 IoU on ∼11-min runs.
- Winning IoU 0.626 came from a weighted blend of three members' models; shipped a TorchScript + FastAPI demo.

HERE Turn-Restriction Validation Challenge May 2026
Team of 3

- Built an XGBoost classifier separating real from erroneous map turn restrictions from GPS probe traces, with GeoPandas/Shapely features in the Belgian national grid; 0.81 F1 under stratified 5-fold CV.
- Generated zero-match synthetic replicas of five proprietary HERE datasets for safe collaboration, and excluded synthetic training sets where labels leaked into route-derived violation features.

LANGUAGES

English (C2) · German (B1–B2) · Russian (B2) · Dutch (A2)